

Hospital Management SQL Assignment

1. Total Number of Patients

Q. Write an SQL query to find the total number of patients across all hospitals.

ANSWER:

```
SELECT SUM("patients_count") AS total_patients
FROM hospital_data;
```

2. Average Number of Doctors per Hospital

Q. Retrieve the average count of doctors available in each hospital.

ANSWER:

```
SELECT
    "hospital_name",
    AVG("doctors_count") AS avg_doctors
FROM hospital_data
GROUP BY "hospital_name";
```

3. Top 3 Departments with the Highest Number of Patients

Q. Find the top 3 hospital departments that have the highest number of patients.

ANSWER :

```
SELECT
    "department",
    SUM("patients_count") AS total_patients
FROM hospital_data
GROUP BY "department"
ORDER BY total_patients DESC
LIMIT 3;
```

4. Hospital with the Maximum Medical Expenses

Q. Identify the hospital that recorded the highest medical expenses.

ANSWER:

```
SELECT
    "hospital_name",
    SUM("medical_expenses") AS total_medical_expenses
FROM hospital_data
GROUP BY "hospital_name"
ORDER BY total_medical_expenses DESC
LIMIT 1;
```

5. Daily Average Medical Expenses

Q. Calculate the average medical expenses per day for each hospital.

ANSWER:

```
SELECT
    hospital_name, admission_date,
    AVG("medical_expenses") AS avg_daily_expenses
FROM hospital_data
GROUP BY hospital_name, admission_date
ORDER BY hospital_name, admission_date;
```

6. Longest Hospital Stay

Q. Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

QUERY:

```
SELECT
    hospital_name,
    department,
    admission_date,
    discharge_date,
    (discharge_date - admission_date) AS stay_days
FROM hospital_data
ORDER BY stay_days DESC
LIMIT 1;
```

7. Total Patients Treated Per City

Q. Count the total number of patients treated in each city.

QUERY:

```
SELECT
    location,
    SUM(patients_count) AS total_patients
FROM hospital_data
GROUP BY location
ORDER BY total_patients DESC;
```

8. Average Length of Stay Per Department

Q. Calculate the average number of days patients spend in each department.

QUERY:

```
SELECT
    department,
    AVG(discharge_date - admission_date) AS avg_stay_days
FROM hospital_data
GROUP BY department
ORDER BY avg_stay_days DESC;
```

9. Department with the Lowest Number of Patients

Q. Find the department with the least number of patients.

QUERY:

```
SELECT
    department,
    SUM(patients_count) AS total_patients
FROM hospital_data
GROUP BY department
ORDER BY total_patients ASC
LIMIT 1;
```

10. Monthly Medical Expenses Report

Q. Group the data by month and calculate the total medical expenses for each month.

QUERY:

```
SELECT
    EXTRACT(MONTH FROM admission_date) AS month,
    SUM(medical_expenses) AS total_medical_expenses
FROM hospital_data
GROUP BY EXTRACT(MONTH FROM admission_date)
ORDER BY month;
```